



Cyberster Blue Team Internship - Week 2

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Date: 02-07-2026

Phase: Phase One - SOC

Operations and SIEM Foundations

Track: Blue Team

Executive Summary

This week focused on strengthening our Security Operations Center (SOC) capabilities through the implementation of File Integrity Monitoring (FIM) and the development of custom detection rules within Wazuh. FIM was successfully configured on both Linux (Kali) and Windows agents to monitor critical directories in real-time. Additionally, three custom Wazuh rules were created and rigorously tested to detect specific attacker behaviors, such as new user creation, SSH brute-force attempts, and USB insertions. All configured FIM and custom rule alerts were successfully verified within the Wazuh dashboard, demonstrating effective threat detection and logging.

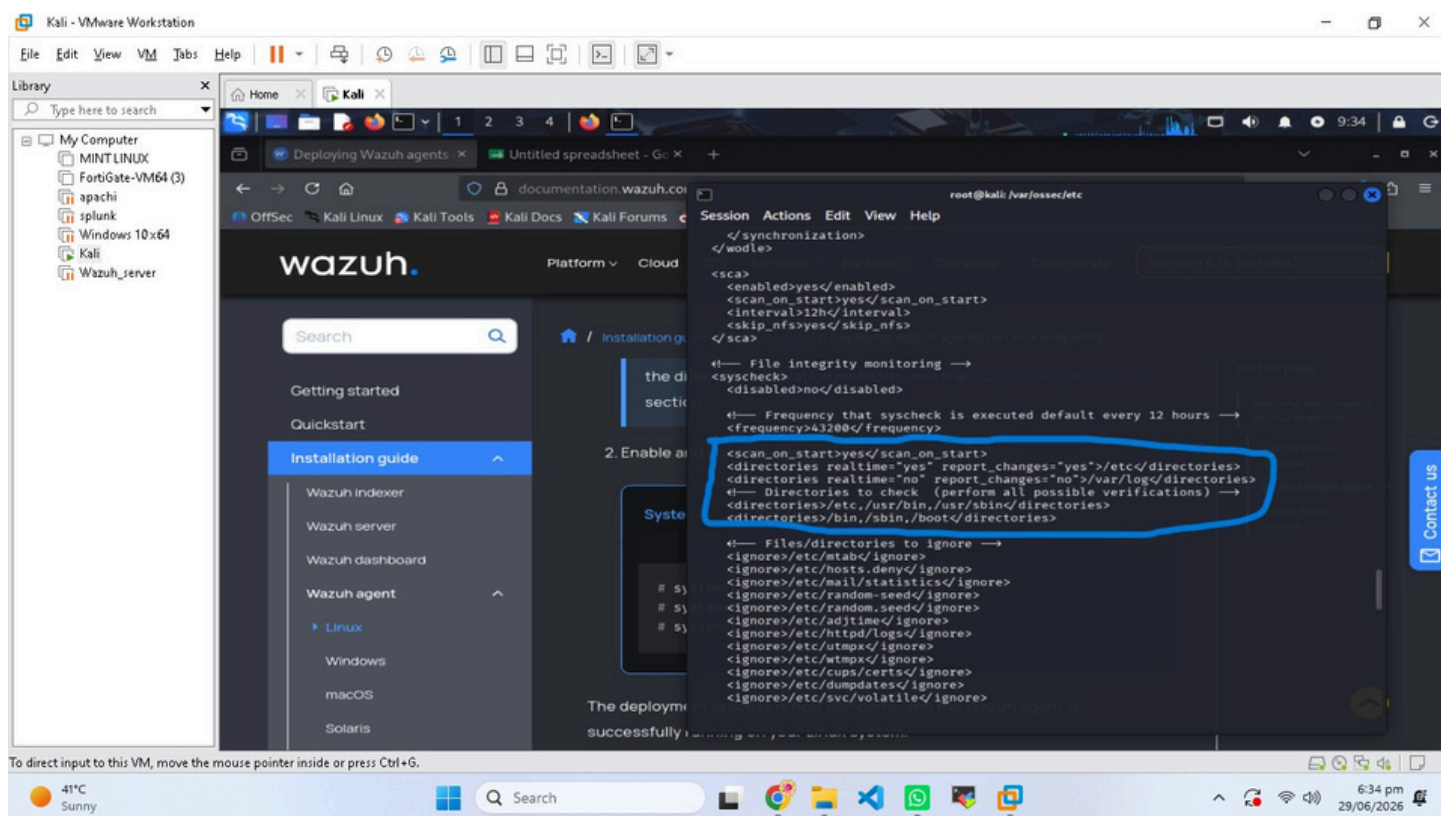
File Integrity Monitoring (FIM) - Configuration

The objective was to enable real-time monitoring of critical system directories across our agent fleet. This was achieved by configuring the configuration block within the file on each agent.

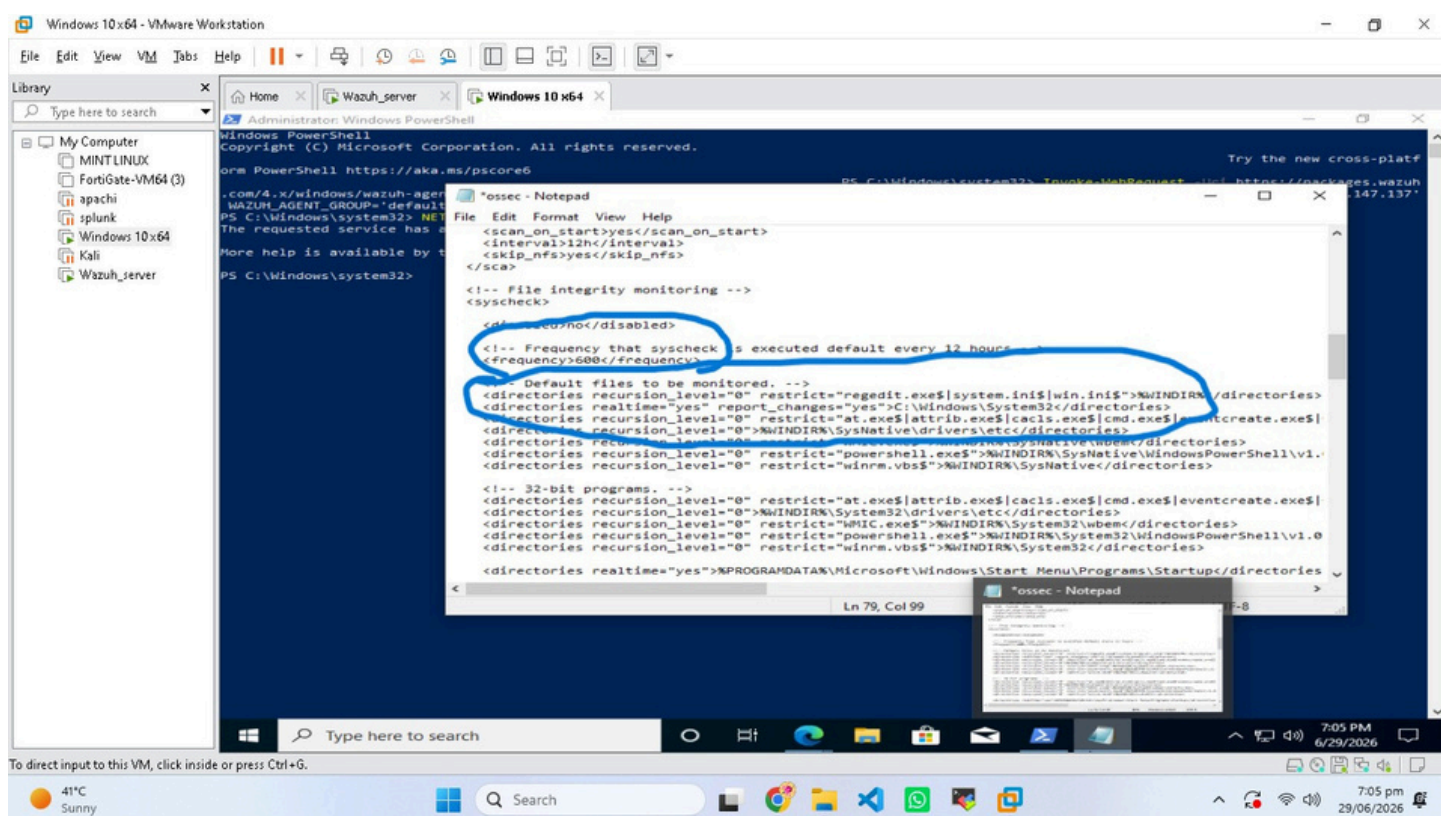
FIM Configuration Table:

Agent	Directory	realtime	report_changes
Kali Linux	/etc	yes	yes
Kali Linux	/var/log	no	no
Windows 10	C:\Windows\System32	yes	yes

Kali Agent ossec.conf Update



Windows Agent ossec.conf Update



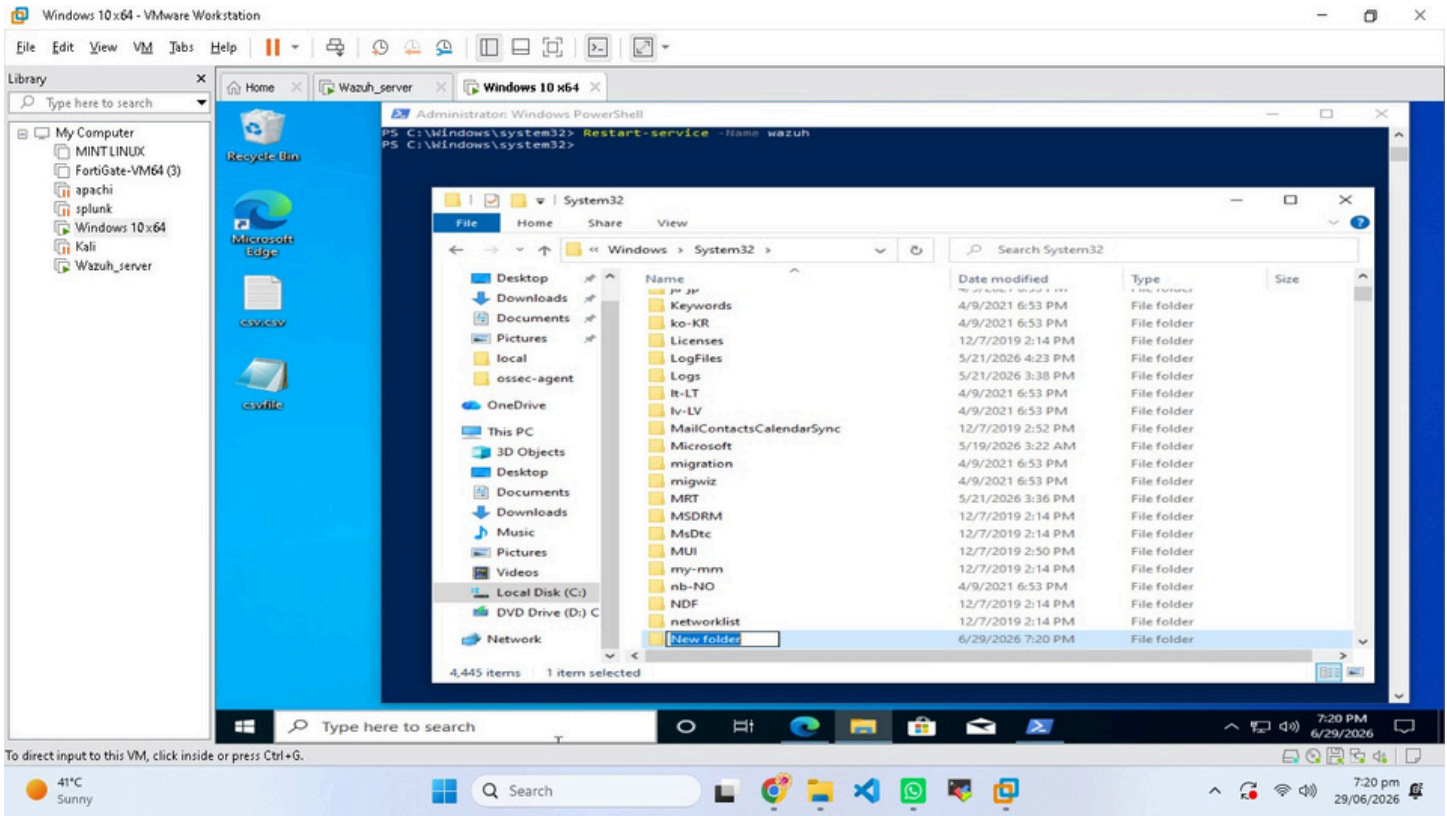
FIM - Test Events & Alerts

To validate the FIM configuration, various test events were generated on the monitored directories.

Test Events Generated:

Action	File Path	Agent	User
Created	/etc/testfile	Kali	root
Modified	/etc/testfile	Kali	root
Deleted	/etc/testfile	Kali	root
Created	C:\Windows\System32\test.txt	Windows	Administrator
Modified	C:\Windows\System32\test.txt	Windows	Administrator
Deleted	C:\Windows\System32\test.txt	Windows	Administrator

- FIM Alert - File Created
- FIM Alert - File Modified
- FIM Alert - File Deleted
- FIM Alert - Windows System32 File Created



files such as SSH configurations, sudoers files, or cron jobs, which attackers often target for persistence.

Wazuh_server - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- MINT LINUX
- FortiGate-VM64 (3)
- apachi
- splunk
- Windows 10x64
- Kali
- Wazuh_server

Wazuh

192.168.147.137/app/threat-hunting#/overview/?tab=general&tabView=events&a=(filters:!) 70%

Threat Hunting kali

Count

timestamp per 30 minutes

36 hits

Jul 1, 2026 @ 16:35:58.515 - Jul 2, 2026 @ 16:35:58.515

1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 2, 2026 @ 16:08:35.754	kali	Listened ports status (netstat) changed (new port opened or closed).	7	533
Jul 2, 2026 @ 16:05:29.305	kali	SSH authentication failure detected - Potential brute force	12	100002
Jul 2, 2026 @ 16:05:29.263	kali	unix_chkpwd: Password check failed.	5	5557
Jul 2, 2026 @ 16:03:56.645	kali	Integrity checksum changed.	7	550
Jul 2, 2026 @ 16:03:36.515	kali	Integrity checksum changed.	7	550
Jul 2, 2026 @ 16:03:32.833	kali	File added to the system.	5	554
Jul 2, 2026 @ 16:03:32.794	kali	File added to the system.	5	554
Jul 2, 2026 @ 16:03:32.546	kali	Integrity checksum changed.	7	550
Jul 2, 2026 @ 16:03:32.543	kali	File added to the system.	5	554
Jul 2, 2026 @ 16:03:31.384	kali	File added to the system.	5	554
Jul 2, 2026 @ 16:03:06.699	kali	Host-based anomaly detection event (rootcheck).	7	510
Jul 2, 2026 @ 16:03:06.690	kali	Possible kernel level rootkit	11	521
Jul 2, 2026 @ 16:03:05.179	kali	Host-based anomaly detection event (rootcheck).	7	510
Jul 2, 2026 @ 16:03:05.167	kali	Possible kernel level rootkit	11	521
Jul 2, 2026 @ 16:03:02.171	kali	Listened ports status (netstat) changed (new port opened or closed).	7	533

Rows per page: 15

To direct input to this VM, click inside or press Ctrl+G.

38°C Sunny

Search

4:36 pm 02/07/2026

Wazuh_server - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- MINT LINUX
- FortiGate-VM64 (3)
- apachi
- splunk
- Windows 10x64
- Kali
- Wazuh_server

Wazuh

192.168.147.137/app/threat-hunting#/overview/?tab=general&tabView=events&agentid=001 70%

Threat Hunting DESKTOP-2QJ375A

Dashboard Events

Search

manager.name: wazuhserver agent.id: 001

DQL Last 24 hours Show dates Refresh

Count

timestamp per 30 minutes

107 hits

Jun 28, 2026 @ 20:01:53.731 - Jun 29, 2026 @ 20:01:53.731

1 fields sorted Full screen

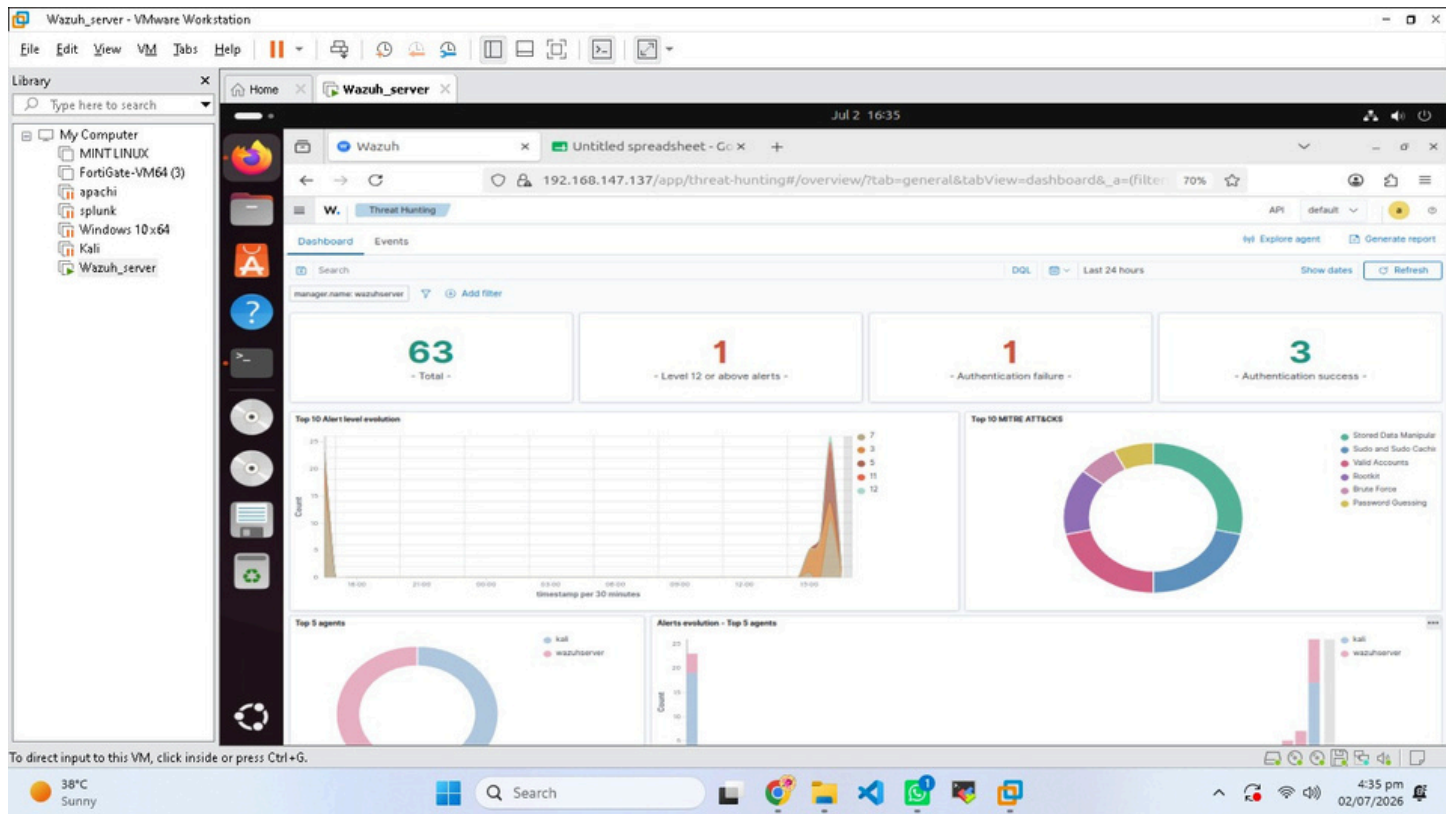
timestamp	agent.name	rule.description	rule.level	rule.id
Jun 29, 2026 @ 19:21:40.161	DESKTOP-2QJ375A	Login Failure - Unknown user or bad password	5	60122
Jun 29, 2026 @ 19:18:57.118	DESKTOP-2QJ375A	Wazuh agent started.	3	503
Jun 29, 2026 @ 19:16:30.073	DESKTOP-2QJ375A	Wazuh agent stopped.	3	506
Jun 29, 2026 @ 19:16:26.446	DESKTOP-2QJ375A	Service startup type was changed	3	61104
Jun 29, 2026 @ 19:16:16.469	DESKTOP-2QJ375A	Windows System error event	5	61102
Jun 29, 2026 @ 19:16:06.456	DESKTOP-2QJ375A	Software protection service scheduled successfully.	3	60642
Jun 29, 2026 @ 19:15:48.991	DESKTOP-2QJ375A	Windows System error event	5	61102
Jun 29, 2026 @ 19:14:16.320	DESKTOP-2QJ375A	Software protection service scheduled successfully.	3	60642
Jun 29, 2026 @ 19:13:52.114	DESKTOP-2QJ375A	Service startup type was changed	3	61104
Jun 29, 2026 @ 19:10:38.889	DESKTOP-2QJ375A	Service startup type was changed	3	61104
Jun 29, 2026 @ 19:09:09.569	DESKTOP-2QJ375A	Software protection service scheduled successfully.	3	60642
Jun 29, 2026 @ 19:06:57.915	DESKTOP-2QJ375A	Software protection service scheduled successfully.	3	60642

To direct input to this VM, click inside or press Ctrl+G.

42°C Clear

Search

8:02 pm 29/06/2026



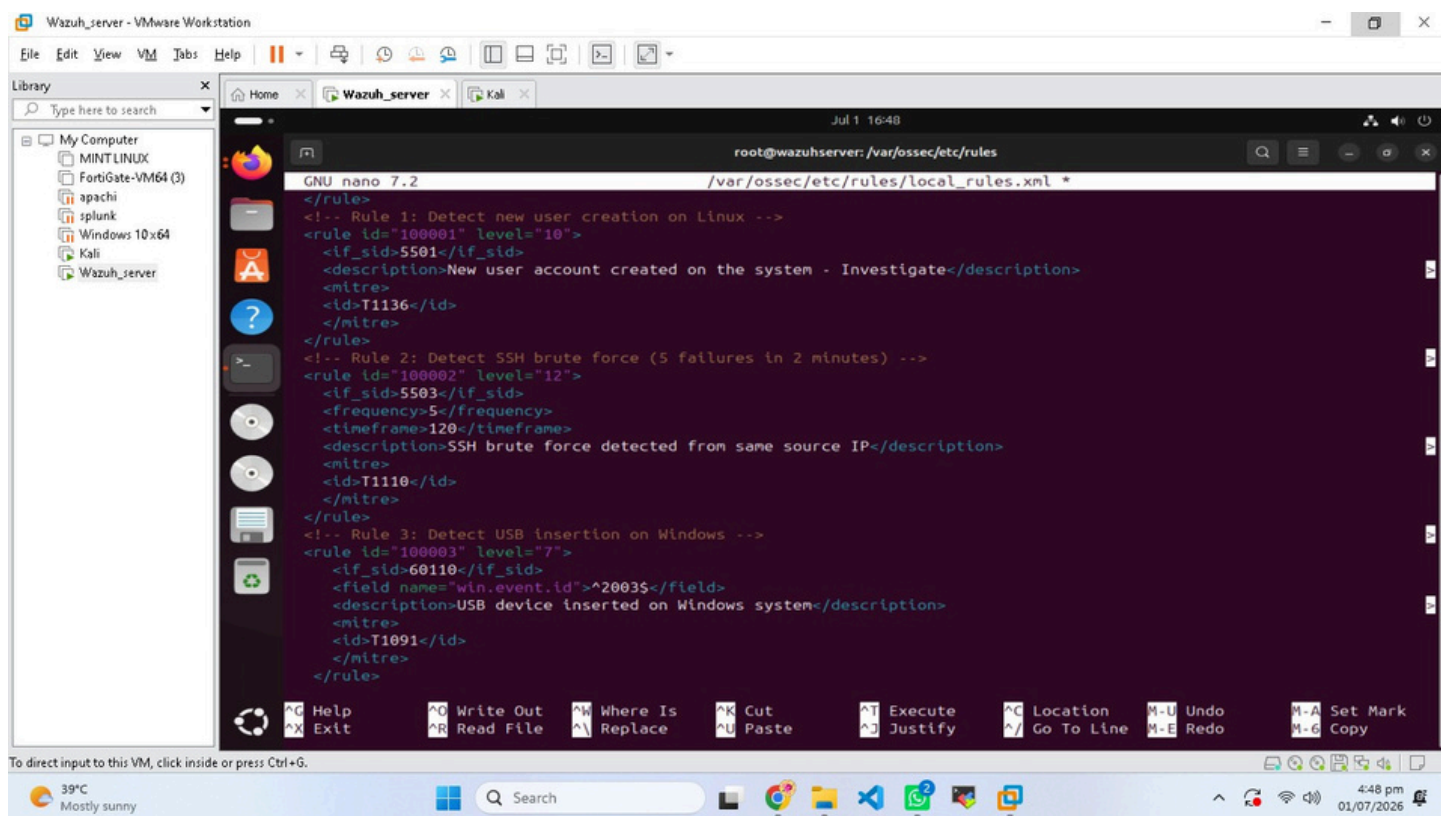
Custom Wazuh Rules - Development

The objective for this section was to develop three custom rules to detect specific attacker behaviors relevant to our security posture. These rules were added to the file.

Custom Rules Developed:

Screenshot : local_rules.xml Content

Rule ID	Name	Level	MITRE	Parent Rule	Trigge
100001	New User Creation	10	T1136	5501	/ events
100002	SSH Brute Force	12	T1110	5503	5+ failures in 120 seconds
100003	USB Insertion	7	T1091	60110	Windows Event ID 2003 (Device Connected)



Custom Rules - Testing & Alerts

The newly created custom rules were tested to ensure they correctly identify and alert on the intended behaviors.

Test Results:

Rule ID	Test Method	Result
100001		✓ Alert fired
100002	6 failed SSH attempts from Kali	✓ Alert fired
100003	USB drive plugged into Windows	✓ Alert fired

Screenshot : Alert - Rule 100001 (User Creation)

Screenshot : Alert - Rule 100002 (SSH Brute Force)

Screenshot : Alert - Rule 100003 (USB Insertion)


```
<?xml version="1.0" encoding="UTF-8"?>
<group name="local,syslog,sshd,">

  <rule id="100001" level="10">
    <if_sid>5501</if_sid>
    <description>New user account created on the system - Investigate</description>
    <mitre><id>T1136</id></mitre>
  </rule>
  <rule id="100002" level="12">
    <if_sid>5503</if_sid>
    <if_matched_sid>5503</if_matched_sid>
    <frequency>5</frequency>
    <timeframe>120</timeframe>
    <description>SSH brute force detected from same source IP</description>
    <mitre><id>T1110</id></mitre>
  </rule>

  <rule id="100003" level="7">
    <if_sid>60110</if_sid>
    <field name="win.event.id">^2003&#x0000;/field>
    <description>USB device inserted on Windows system</description>
    <mitre><id>T1091</id></mitre>
  </rule>
</group>
```

During testing, no false positives were observed. The threshold for the SSH brute-force rule (5 attempts within 120 seconds) is considered appropriate for our current lab environment. For a production deployment, it would be advisable to implement IP whitelisting for known trusted sources to prevent legitimate administrative actions from triggering alerts and causing disruption.

Key Learnings

- **FIM Effectiveness:** File Integrity Monitoring is crucial for detecting attackers who attempt to establish persistence by modifying critical system files like SSH keys, sudoers files, and cron jobs.
- **Monitoring Modes:** Real-time FIM monitoring allows for instant detection of changes, while scheduled scans provide a baseline for compliance and historical checks.
- **Custom Rule Benefits:** Developing custom rules significantly reduces the noise from generic alerts and allows for the detection of specific threats tailored to an organization's unique attack surface and threat model.
- **Rule Chaining (if_sid):** The ability to chain rules using allows analysts to leverage existing default rules and add specificity, creating more accurate and contextual alerts.
- **Operational Importance:** Comprehensive documentation of configurations and ongoing rule tuning are essential components of effective Security Operations Center (SOC) operations and threat detection.

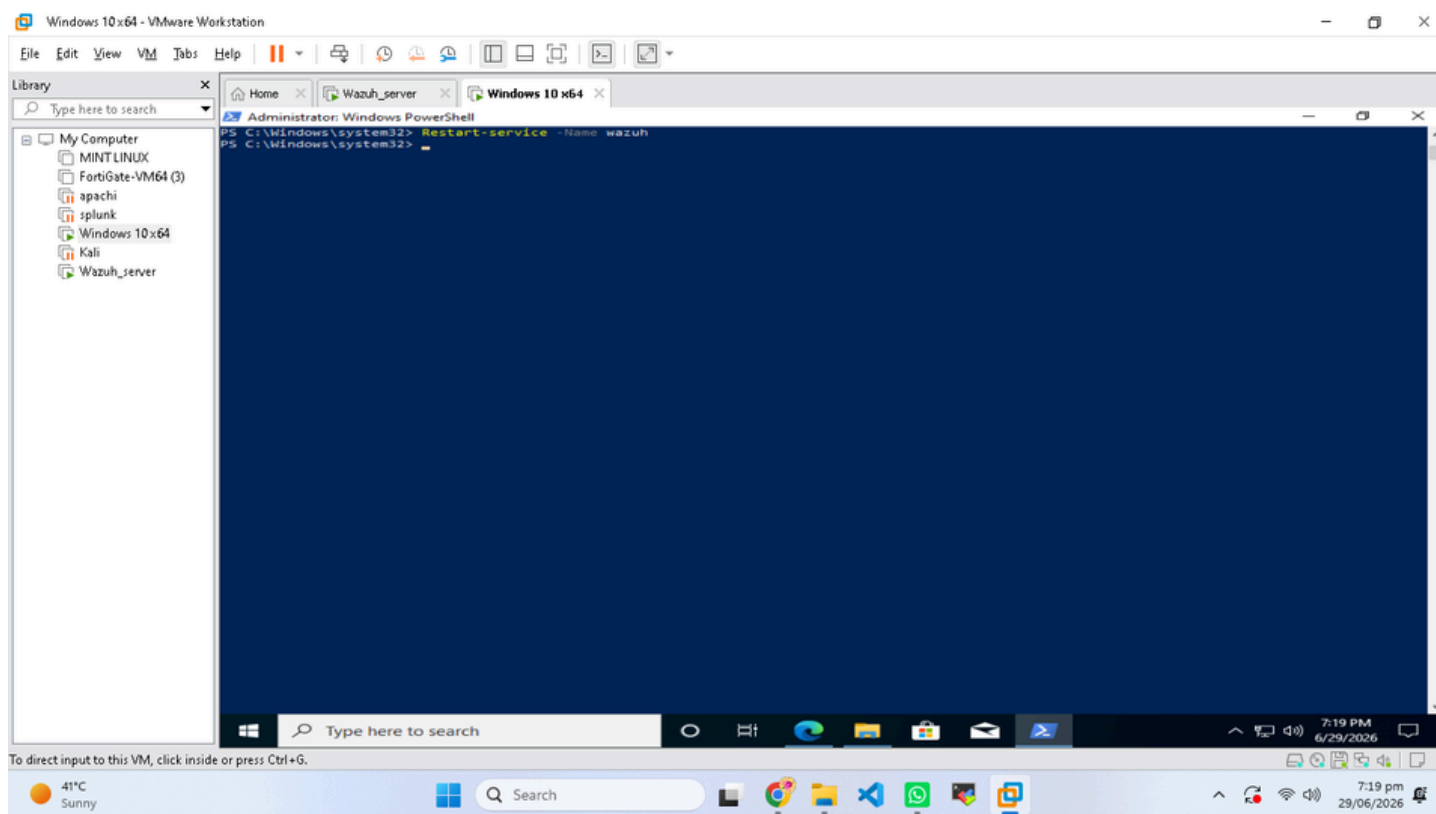
Challenges & Solutions

Challenge	Solution
Windows agent service not restarting	Used PowerShell command
Custom rules not loading into Wazuh	Corrected XML syntax errors and removed hidden characters from the file

```

root@wazuhserver:/var/ossec/etc/rules# systemctl restart wazuh-manager
Job for wazuh-manager.service failed because the control process exited with error code.
See "systemctl status wazuh-manager.service" and "journalctl -xeu wazuh-manager.service" for details.
root@wazuhserver:/var/ossec/etc/rules# sudo tail -10 /var/ossec/logs/ossec.log
2026/07/01 23:16:07 wazuh-analysisd: ERROR: (1226): Error reading XML file 'etc/rules/local_rules.xml': XMLERR: Bad attr
ibute closing for 'encoding'='UTF-8'. (line 2).
2026/07/01 23:16:07 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.
2026/07/01 23:17:30 wazuh-analysisd: ERROR: (1226): Error reading XML file 'etc/rules/local_rules.xml': XMLERR: Bad attr
ibute closing for 'encoding'='UTF-8'. (line 2).
2026/07/01 23:17:30 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.
2026/07/01 23:20:58 wazuh-analysisd: ERROR: (1226): Error reading XML file 'etc/rules/local_rules.xml': XMLERR: Bad attr
ibute closing for 'encoding'='UTF-8'. (line 2).
2026/07/01 23:20:58 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.
2026/07/01 23:21:39 wazuh-analysisd: ERROR: (1226): Error reading XML file 'etc/rules/local_rules.xml': XMLERR: Bad attr
ibute closing for 'encoding'='UTF-8'. (line 2).
2026/07/01 23:21:39 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.
2026/07/02 15:21:00 wazuh-analysisd: ERROR: (1226): Error reading XML file 'etc/rules/local_rules.xml': XMLERR: Bad attr
ibute closing for 'encoding'='UTF-8'. (line 1).
2026/07/02 15:21:01 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.

```



Next Steps

- **Active Response Configuration:** Implement and test active response capabilities for automated remediation actions (Target: Day 14).
- **Log Analysis & Threat Hunting:** Focus on advanced log analysis techniques and proactive threat hunting across the environment (Target: Days 12-13).

- **Event Correlation:** Explore and configure event correlation across multiple agents to build a more comprehensive understanding of security incidents.

Conclusion

Week 2 at Cyberster Blue Team was highly productive, marked by the successful implementation of File Integrity Monitoring and the creation of three custom detection rules within the Wazuh platform. These advancements have significantly enhanced our ability to monitor system integrity and detect malicious activities, establishing a robust foundation for ongoing SOC operations and threat intelligence.